



Artificial intelligence driven cyberattack detection system using integration of deep belief network with convolution neural network on industrial IoT

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ARTICLE INFO

Keywords:

Industrial Internet of Things
Cybersecurity
Honey Badger Algorithm
Ensemble Learning
Attack Detection

ABSTRACT

In Industry 4.0, information and communication technology (ICT) was employed in numerous significant infrastructures, like financial networks, smart factories, and power plants, to automate and certify industrial systems. In power control systems, ICT technologies such as IIoT have improved automated monitoring, but legacy methods, originally autonomous, now connect with external networks. This progress has presented safety vulnerabilities from legacy ICT systems. Hence, various cybersecurity approaches are developed and examined to deal with cyberattacks and vulnerabilities. Utilizing new cybersecurity models in power control systems poses risks due to their uncertified safety. Ensuring their stability and efficiency is significant for maintaining reliable power delivery and incorporating these technologies into power control systems. Therefore, this study designs a Next-Generation Cybersecurity Attack Detection using an ensemble deep learning model (NGCAD-EDLM) technique in the IIoT environment. The main cause of the NGCAD-EDLM technique is the automatic recognition of cyber-attacks. In the NGCAD-EDLM approach, the primary data normalization phase utilizing min-max normalization is performed. Next, the honey-badger algorithm (HBA) approach selects the feature subsets. Furthermore, an ensemble deep learning (DL) of two methods, namely convolutional neural networks (CNNs) and deep belief networks (DBNs) methods, are employed for classification. In addition, the DL techniques' hyperparameter selection is accomplished using the lotus effect optimization algorithm (LEOA) method. A complete set of simulation validation is performed to establish the experimental analysis of the NGCAD-EDLM method. The performance validation of the NGCAD-EDLM method exhibited a superior accuracy value of 99.21 % over other existing techniques.

1. Introduction

The Industrial Internet of Things (IIoT) can be the expansion of the classic IoT uses during the industrialized sector [1]. The IIoT improves industries' abilities to offer enhanced productivity and reliability in industrialized operations. In a smart manufacturing system, a

manufacturing system's observing and controller abilities are considerably enhanced by combining all the other cyber-physical methods and new communiqué techniques [2]. The theory of smart industrialization is most significant in recognizing the idea of the next generation of the industrialized uprising, which is considered Industry 4.0. An outstanding amount of actuators, innovative technologies, and sensors

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<https://doi.org/10.1016/j.aej.2024.10.009>

Received 29 July 2024; Received in revised form 14 September 2024; Accepted 2 October 2024

Available online 14 October 2024

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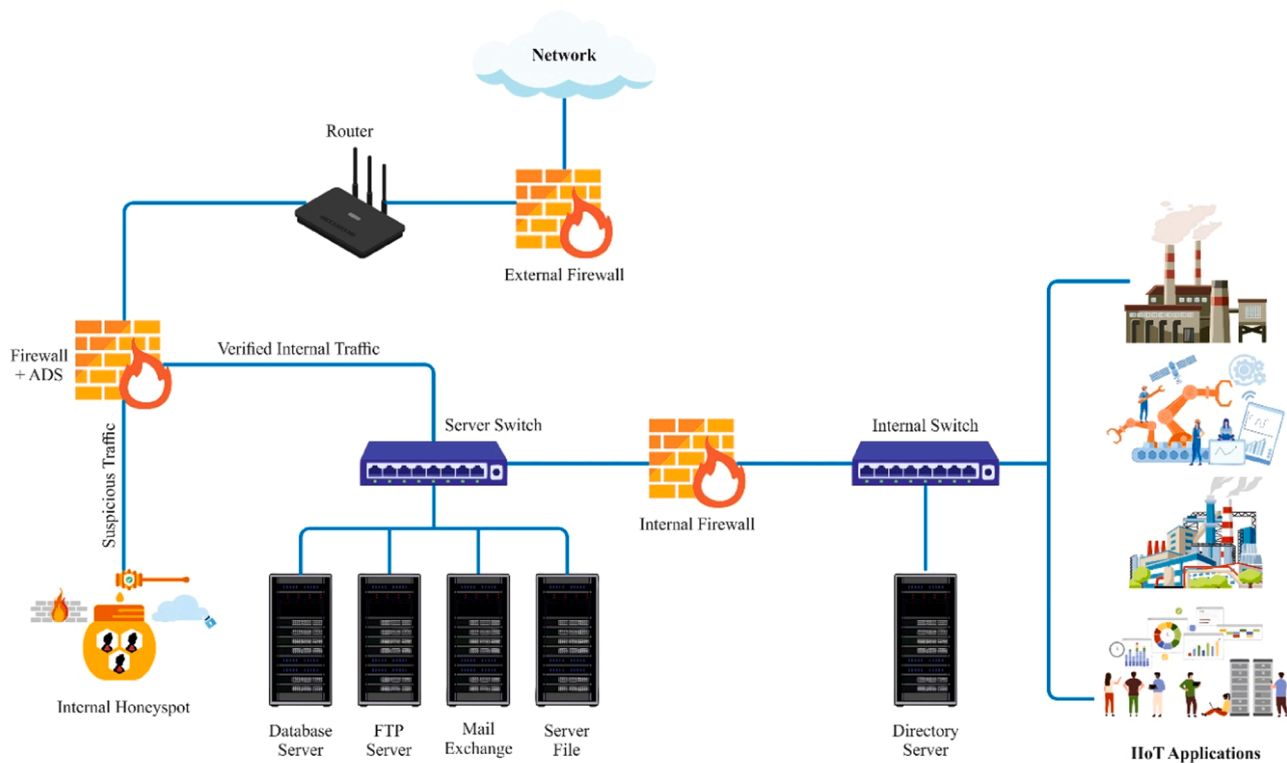


Fig. 1. General architecture of internal cybersecurity.

are incorporated into the manufacturing sector [3]. Network latency, response time, and reliability are significant reasons in the modern industry. With all these factors, decision-making and data transmission technology are optimized without human interaction. Based on the vision of Industry 4.0, the application of the IoT among manufacturing sectors increases the security, effectiveness, and production of industrialized processes [4]. Concisely, the IIoT is primarily related to the effective usage of the IoT in the industrial area. Fig. 1 illustrates the general structure of internal cybersecurity. However, with the fast growth among several IIoT edge devices within industries, some security and secrecy difficulties have risen, which is a great task for the honesty and security of the IIoT [5]. The edge device was a utilization point for intruders. The co-operated IIoT devices could be a reason for transferring false data into the cloud server or result in unapproved admission to valued corporate data, professional plans, and industrial documents. These might cause operating ineffectiveness and reputational and financial loss [6].

Safeguarding cybersecurity in the IIoT is the foremost task in current manufacturing atmospheres. It contains the safety on edge devices of illegal access and malware and guarantees physical and communication privacy. The reliability and security privacy of the IIoT may be enhanced by using robust and improved security tools. Concerning IoT or IIoT security, an intrusion detection system (IDS) is a broadly approved methodology. An IDS is utilized to monitor networks for policy violations or malicious activity. Artificial intelligence (AI) is applied to security methods, such as decreasing cyber security risks. Now, a machine learns from the input data and creates an upcoming prediction [7]. It is applied in email filters to deal with online search engines, websites for creating personalized recommendations, spam, banking software to detect rare transactions, and various apps on phones such as voice recognition. For cybersecurity, ML became an important technique. Using ML, the patterns are examined by cybersecurity and learned with them to check related attacks and react to varying behaviour [8]. Nowadays, the usage of deep learning (DL) models to develop and design cyber-security results has gained considerable attention from the

industry and the academic world. The DL methods need important opportunities to produce enhanced outcomes from the industrialized method of big data. Still, the proposal of robust and better attack recognition systems for the IIoT is silent in open tasks. The rapid evolution of industrial systems calls for advanced solutions to address growing safety threats [9]. As industrial environments adopt advanced technologies and interrelated devices, conventional safety measures become inadequate. Improved protection mechanisms are significant to safeguard these critical systems from intrinsic cyber-attacks. More robust and adaptive defence strategies can be deployed by employing cutting-edge methods in machine learning and artificial intelligence. This methodology fortifies the integrity of industrial operations and confirms that productivity and reliability remain uncompromised in the face of growing cyber threats [10].

This study designs a next-generation Cybersecurity Attack Detection utilizing an ensemble deep learning model (NGCAD-EDLM) method in the IIoT environment. The main cause of the NGCAD-EDLM method rests in the automated recognition of cyber-attacks. In the NGCAD-EDLM approach, the main data normalization phase utilizing min-max normalization is performed. Next, the honey-badger algorithm (HBA) approach is used to select feature subsets. Furthermore, an ensemble deep learning (DL) model of two methods, namely convolutional neural networks (CNNs) and deep belief networks (DBNs), are employed for classification. In addition, the DL techniques' hyperparameter selection is accomplished using the lotus effect optimization algorithm (LEOA) method. A complete set of simulation validation has been conducted to establish the experimental analysis of the NGCAD-EDLM method. The major contribution of the NGCAD-EDLM method is listed below.

- The NGCAD-EDLM approach incorporates min-max normalization to standardize feature values, which confirms that the input data is scaled appropriately for enhanced performance. This methodology improves the technique's capability to handle varied data ranges effectively. By standardizing the data, the model achieves more precise and reliable outputs.

- The NGCAD-EDLM technique employs the HBA model to choose the most relevant feature subsets, substantially improving efficiency and accuracy. This methodology prioritizes key features, resulting in more precise and efficient classification. The method attains improved accomplishment and reduced computational complexity by concentrating on essential data.
- The NGCAD-EDLM methodology integrates CNN and DBN techniques in an ensemble approach, providing robust and accurate classification. This incorporation utilizes the merits of both methods to improve comprehensive performance and reliability. By combining CNN and DBN models, the method enhances its capability for handling intrinsic data patterns and attains greater classification outputs.
- The NGCAD-EDLM model utilizes the LEOA method to fine-tune hyperparameters, improving the performance and efficiency of the deep learning approaches. This optimization confirms that the technique attains the optimum possible outcomes by altering key parameters for enhanced accuracy and effectiveness. By utilizing the LEOA technique, the model efficiently optimizes the potential of its deep learning elements.
- The NGCAD-EDLM methodology uniquely incorporates min-max normalization, feature subset selection with HBA, ensemble deep learning models such as CNN and DBN, and hyperparameter optimization through LEOA. This integration creates a comprehensive and effective classification task framework, optimizing data processing and model performance. By incorporating these advanced methods, the technique attains improved accuracy and robustness in handling intrinsic classification threats.

2. Review of literature

Souri et al. [11] propose a novel cloud-based cyberattack recognition structure based on Ensemble Bagged Trees Detection (EBTD) methods for forecasting cruel actions and cyberattack kinds for hyper-automated processes in the IIoT. The presented structure utilizes the extraction model and priority-based FS and Analysis of Variance (ANOVA) to detect the optimum feature with high depending on the malicious behaviours, calculation time, network traffic, and attack kinds. Ajay et al. [12] propose a DBN-protected material-enhanced symmetrical intrusion avoidance sensor model that enhances the safety of IoT devices. DBNs enhance The system's ability to prevent and identify cyberattacks. The study equates the recommended technique's performance to industry normal IDS methods and the domain generation algorithm (DGA) to measure efficiency. Naveeda and Fathima [13] propose an IoT-E-CADS for advanced metering infrastructures (AMIs). Based on industry standards, the recommended Bi-level IoT-E-CADS can recognize two various types of attacks in a smart grid set. The Isolation Forest (IF) method for ML is utilized early to identify cyberattacks and differences in real systems. Afterwards, the DT-ML method is used to recognize cyber-attacks and occurrences of false data injection in real systems. Zanasi et al. [14] propose a new security structure primarily aimed at tackling the stringent necessities of IIoT methods. It is dependent upon a network micro-segmentation that could be continuously combined into current atmospheres, and two major parts: a software-defined network (SDN) ensures a united abstraction layer for policy execution through various atmospheres and a central security organization layers, which simplify the policy implementation of one structural designs.

[15] presents a new Gradient Descent Scaling and Segmented Regression fine-tune Federated Learning (GDS-SRFFL) technique for IIoT network attack recognition. The objective of the GDS-SRFFL technique is to increase the safety of the IIoT network. At first, the innovation of GDS-based pre-processing is employed in the raw datasets to get feature-scale pre-processed network samples. Next, the unnecessary intrusions are identified by utilizing a (GDS-SRFFL) method, ensuring the security of the IoT network by the originality of SoftMax Regression. Khan et al. [16] introduce a new method to improve the reliability of

IIoT-enabled networks. The authors present consistent and precise supervisory control and data acquisition (SCADA) network-based cyber-attack detection in these networks. The suggested system combines the DL-based decision tree (DT) and pyramidal recurrent unit (PRU) by SCADA-based IIoT system. Also, an ensemble learning (EL) technique is utilized to identify cyber-attacks in SCADA-based IIoT networks. The non-linear learning capability of the ensemble DT and PRU addresses the inappropriate feature sensitivity and allows higher recognition rates. Huang et al. [17] introduced Dynamic Watermarking (DW) to protect ICS. The main concept of DW is that it increases a private random signal watermark, then spreads throughout the plant, is properly transformed, and then returns to the attack detectors. The actuation side of the attack detector can identify a Man-in-the-middle (MiTM) or Masquerade attack on the cascading system in real-time.

In [18], a pre-training technique using a deep AE and DNN was presented for fastly forecasting attacks by reduced false rate and increased accuracy. The replicas are extended by utilizing hyper-parameter optimizer (HPO) methods. The presented method provides an alternative to DL construction replicas over an HPO process integrating the Archimedes optimizer method. This optimizer method could define hyperparameter values and integration for enhanced recognition performances. Huma et al. [19] propose a novel hybrid deep random neural network (HDRaNN) technique for cyberattack detection in IIoT. HDRaNN model incorporates a deep random neural network with a multilayer perceptron featuring dropout regularization. Idougli et al. [20] introduce a novel intrusion detection model against growing cyber threats by employing machine learning and deep learning methods for dynamic adaptation. Islam et al. [21] present an IDS for Industrial Cyber-Physical Systems (ICPS) using Generative AI and cognitive computing. It incorporates a Long Short-Term Memory-based Sparse Variational Autoencoder (LSTM-SVAE) method for feature extraction, a Bidirectional Recurrent Neural Network with Hierarchical Attention (BiRNN-HAID) approach for intrusion detection, and Cognitive Enhancement for Contextual Intrusion Awareness (CE-CIA) approach to refine predictions. The system also comprises Interpretive Assurance through Activation Insights in Detection Models (IAA-IDM) methodology for visualizing layer activations to support cybersecurity analysis.

Kilichev, Turimov, and Kim [22] propose an innovative intrusion detection approach for IoT-based electric vehicle charging stations (EVCS), incorporating CNNs, LSTM, and gated recurrent unit (GRU) models. Hamouda et al. [23] introduce FedGenID, a Federated Generative Intrusion Detection System for IIoT networks, featuring a three-model framework: a federated generative model with cGANs for data augmentation, a secondary model for improving attack detection, and a model for precise threat detection using augmented data. FedGenID efficiently addresses non-IID, multi-class cyber threats and illustrates resilience to adversarial attacks. Bakhsh et al. [24] present a deep learning-based IDS for IoT networks using Feed Forward Neural Networks (FFNN), LSTM, and Random Neural Networks (RandNN). FFNN manages intrinsic traffic patterns, LSTM captures long-term dependencies, and RandNN adapts with flexible dynamics. Together, these methods improve cybersecurity by efficiently defending against cyber threats and safeguarding sensitive data in expanding IoT networks. Gaurav et al. [25] propose a deep learning framework for safe IoT and cloud computing integration, improving cyber defence. It employs optimized feature selection and a CNN technique to effectually filter attack traffic and tackle critical safety threats in IoT-cloud systems. Ehmer et al. [26] introduce a shallow neural network encompassing only 110 ReLU-activated artificial neurons capable of detecting representative attacks seen on a communication network.

The presented methods in recent studies illustrate various limitations. The EBTD methodology may be computationally intensive and intrinsic, potentially hindering real-time application and feature selection. While efficient, the DBN-based method might face difficulty with scalability and various attack scenarios. The early-stage detection

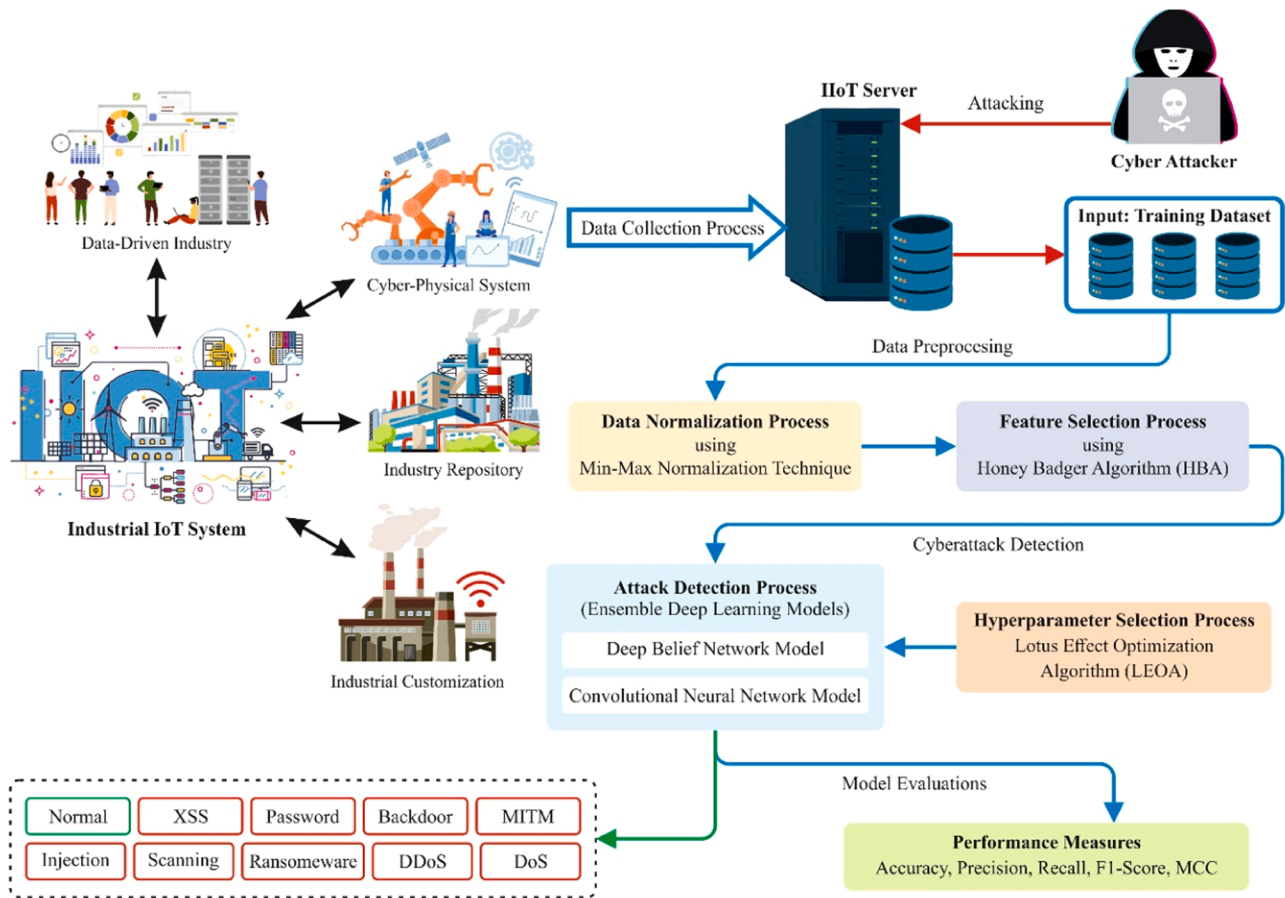


Fig. 2. Workflow of NGCAD-EDLM approach.

technique utilizing IF may miss state-of-the-art attacks, and its focus on false data injection might need to address other attack kinds. Network micro-segmentation and software-defined network approaches could be complex to incorporate and may only cover some safety requirements. The GDS-SRFFL model might not handle all network traffic types efficiently, and pre-processing steps could present biases. The EL technique might still encounter threats with complex attacks, and DW could add processing overhead and may need to address the overall attack kinds. Finally, hyperparameter optimization techniques and pre-training with deep autoencoders may involve significant computational resources and may not always yield optimal configurations. Despite advances in cybersecurity for IIoT systems, existing methods often struggle with real-time performance, scalability, and adaptability to diverse attack types. There is a need for more efficient, versatile approaches that can seamlessly integrate into various industrial environments and address emerging, sophisticated threats while minimizing computational overhead.

3. Methodology

In this study, an NGCAD-EDLM technique is proposed in the IIoT environment. The major cause of the NGCAD-EDLM technique rests in the automatic recognition of cyber-attacks. The NGCAD-EDLM approach has data normalization, HBA-based feature selection, EL, and parameter optimizer to accomplish that. Fig. 2 depicts the workflow of the NGCAD-EDLM approach.

3.1. Min-max normalization

Initially, the NGCAD-EDLM approach undergoes data normalization

utilizing min-max normalization has been carried out. The min-max normalization is vital in cybersecurity to detect attacks in IIoT environments [27]. By scaling data to a permanent range, generally [0,1], min-max normalization certifies that features with dissimilar scales and units donate similarly to the analysis. This is mainly significant in IIoT, where dissimilar sensors produce heterogeneous data streams. Normalized data improves the ML method's performance by enhancing the accuracy of anomaly recognition and decreasing false positives. As a result, it helps in the on-time identification and minimization of cyberattacks, establishing the security and robustness of IIoT systems.

3.2. Feature selection process

Next, the HBA approach is utilized to select the optimum feature subsets. The HBA method is stimulated by the behaviour of honey badgers [28]. The HBA approach is a compelling choice for optimal feature subset selection due to its robust performance in handling intrinsic and high-dimensional datasets. Inspired by the foraging behaviour of honey badgers, the HBA model outperforms in exploring an ample search space to detect the most relevant features with high accuracy. Its capability to effectively navigate numerous feature combinations confirms that only the most crucial features are chosen, enhancing the model's performance and reducing computational complexity. Furthermore, the resilience of the HBA model to local optima and its adaptive search capabilities make it appropriate for dynamic and diverse data environments, improving both the effectualness and efficiency of feature selection processes. Due to its flag-based modifications and dynamic randomization strategies, the HBA method mitigates the risk of local optimum trapping more efficiently than many other optimization models. These features improve its exploration

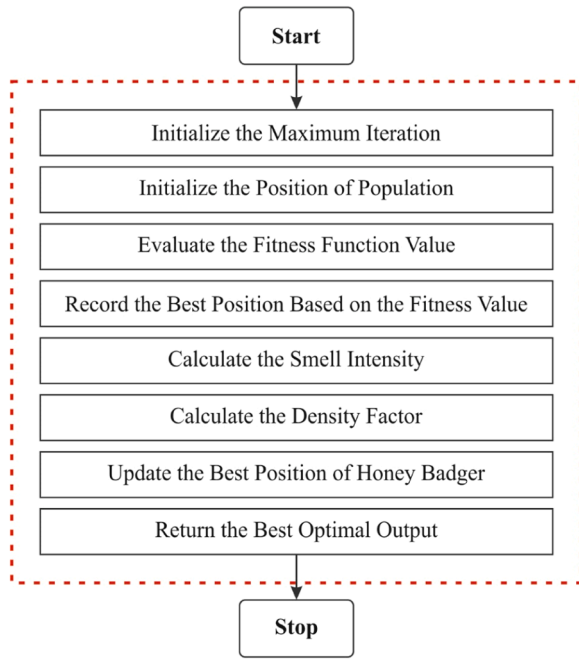


Fig. 3. Steps involved in the HBA method.

capabilities by allowing it to escape local optima and explore a broader search space. The flag-based modifications assist in prioritizing promising regions, while dynamic randomization averts premature convergence, thus enhancing the capability of the technique to find globally optimal solutions across diverse datasets. Fig. 3 illustrates the steps involved in the HBA model.

In general, the honey badger is a bold animal that is mainly seen in Asia and Africa. They usually like beehives and honey, but they cannot identify the position of hives. This issue can be resolved by the badger and honeyguide direction; a bird can discover the hives but can't attain honey, which leads it to beehives. Furthermore, by utilizing sensing skills, the honey badger targets 60 species. It begins by assessing the victim's position and then moves nearby in the victim's neighbourhood to discover an appropriate location to capture the victim. The badger starts catching and digging once the position is found accurately. HBA simulates the behavior of badger feeding in dual methods. At first, the smell, mode, and dig can be recognized as the digging method, and the following method, honeyguide, can be identified as a honey method.

The HBA method begins by setting the search space, which is expressed using Eq. (1).

$$\text{Candidate solutions} = [x_{11} \dots x_{1D} : \dots : x_{n1} \dots x_{nD}] \quad (1)$$

Here, N denotes the number of badgers, and each location is set. Then, compute the intensity (I) spread on the distance and the prey smell. The speed is based on the capability of smell. The smell power is calculated by *Inverse Square Law* and expressed in Eqs. (2), (3), and (4). The r_1 picks at random among 0 and 1, x_i denotes the candidate solution, lb_i and ub_i signify the limits of the search space, whereas the initial part is the lower limit and the second one is the upper limit. I_i refers to the smell intensity of prey, r_2 is nominated at random among 0 and 1, S refers to the location of prey (concentration strength), d_i signifies the distance between prey and badger, and x_{prey} denotes the position of prey. The 3rd stage is upgrading the density factor (α). It certifies the smoothness of the exploration and exploitation stages by monitoring the dynamic randomization. The density factor reduces with period to minimize the randomization as per Eq. (6). While $t_{\max}$ denotes the maximum iteration count and C refers to a constant $= 2$.

The significant phase in meta-heuristic techniques is evading the local optimum trapping. The HBA transforms the search by utilizing a

flag (F). The HBA upgrades the location procedure in dual stages, such as digging and honey. In the stage of digging, the badger upgrades its location by flying like the nature of a Cardioid. This gesture can be pretended by Eq. (7). While x_{new} denotes the badger's novel location, F refers to the direction-modified flag, β epitomizes the capability to acquire food, and r is a number chosen randomly among 0 and 1. The F is computed utilizing Eq. (8). In the stage of honey, the badger upgrades his location by tracking the directed bird, and this gesture is pretended by Eq. (9). Here, r_7 is chosen randomly among 0 and 1.

$$x_i = lb_i + r_1 \times (ub_i - lb_i) \quad (2)$$

$$I_i = \frac{r_2 \times S}{4\pi d_i^2} \quad (3)$$

$$S = (x_i - x_{i+1})^2 \quad (4)$$

$$d_i = x_{prey} - x_i \quad (5)$$

$$\alpha = C \times \exp\left(-\frac{t}{t_{\max}}\right) \quad (6)$$

$$x_{new} = x_{prey} + F \times \beta \times I \times x_{prey} + F \times r_3 \times \alpha \times d_i \times \cos(2\pi r_4) \times [-\cos(2\pi r_5)] \quad (7)$$

$$\text{if } \text{ifr}_6 \leq 0.5 \text{ then } F = -1 \text{ else } F = 1 \quad (8)$$

$$x_{new} = x_{prey} + F \times r_7 \times \alpha \times d_i \quad (9)$$

The fitness function (FF) utilized in the HBA is created to balance the selected feature numbers in every solution (minimum) and the classifier precision (maximum) achieved by using these nominated features. Eq. (10) presents the FF to compute the solution.

$$\text{Fitness} = \alpha \gamma_R(D) + \beta \frac{|R|}{|C|} \quad (10)$$

Whereas $\gamma_R(D)$ presents the classifier error rate for an assumed classifier. $|R|$ denotes the cardinality of the nominated sub-set, and $|C|$ represents the complete count of features in the database; α and β are two parameters equivalent to the significance of classifier subset length and quality. $\in [0,1]$ and $\beta = 1 - \alpha$.

3.3. EL model

Furthermore, an ensemble DL of two methods, namely CNNs and DBNs, are applied for the classification process. Integrating these models in an ensemble DL methodology improves the classification process by implementing the merits of both techniques. The CNN models are highly efficient for extracting spatial features and patterns from data, making them ideal for image and signal processing tasks. On the contrary, DBNs outperform in capturing intrinsic, hierarchical structures in data, which complements the capabilities of the CNN technique. This ensemble methodology enhances classification accuracy by incorporating the capability of the CNN technique to handle spatial hierarchies with DBNs' proficiency in learning deep, abstract features. The synergy between these methods addresses a broader range of data characteristics and enhances the model's overall performance compared to using either technique alone.

3.3.1. CNN classifier

CNN is utilized for different types of data, not just images [29]. In the case of 1D data, such as time series or sequences, 1D convolution is utilized. In this study, 1D-CNN is utilized with $\text{kernel_size} = 2$ for capturing relatively short-term features in the EEG dataset, the max-pooling layer assists in reducing the spatial dimension, and the *ReLU* activation function learns complicated relationships in the data.

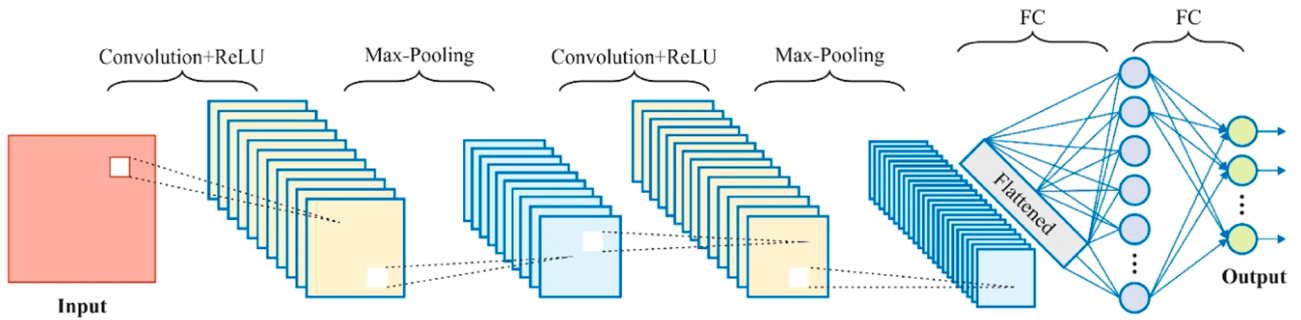


Fig. 4. Framework of CNN model.

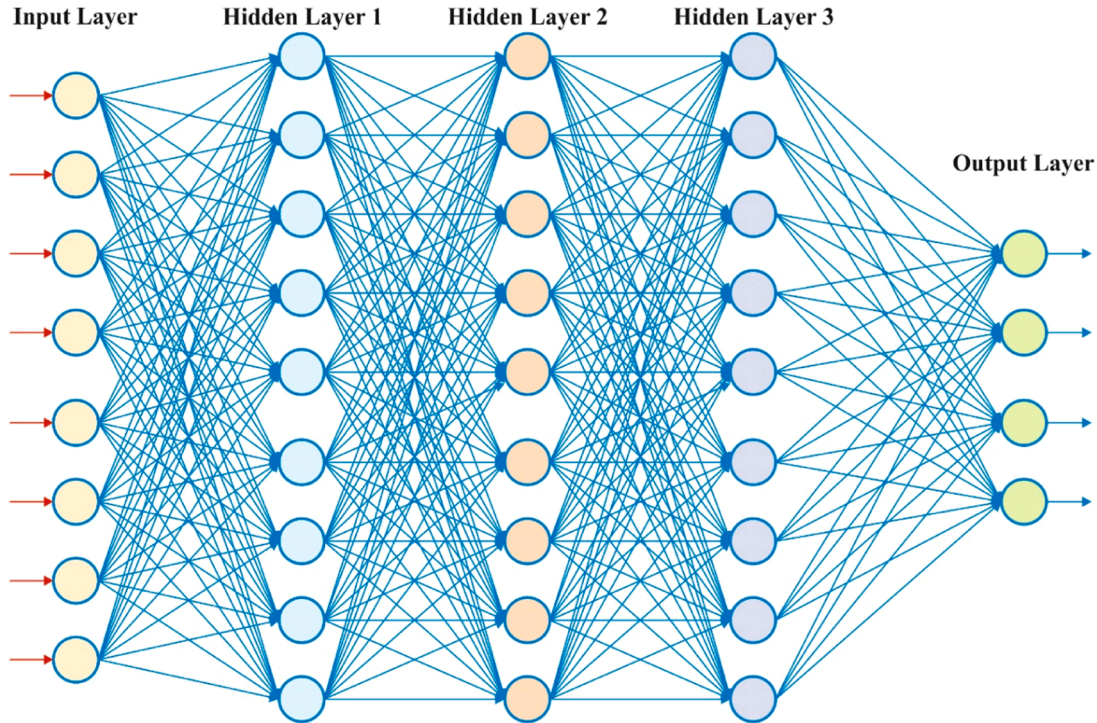


Fig. 5. Structure of the DBN approach.

This model uses four convolution layers with filters of 32, 32, 64, and 128. The gradual increase in the filter across the convolutional layers implies a hierarchical feature learning process. Deep layers with more filters learn high-level and more abstract representations. This enables extracting hierarchical features at dissimilar abstraction levels from the EEG information. In this model, the dropout rate 0.2 shows a regularization strategy to avoid over-fitting. The first FC layer contains 64 neurons with ReLU and a dropout rate 0.5. The last FC layer contains one neuron with a Sigmoid. Now, Adam Optimizer is utilized with the “learning_rate” equivalent to 0.0005 and ‘binary_cross – entropy’ as a loss function. Fig. 4 illustrates the framework of the CNN model.

The below equation represents the mathematical formula of the Sigmoid, ReLU, and Binary cross-entropy functions.

$$\text{Sigmoid}(x) = \frac{1}{1 + e^{-x}} \quad (11)$$

Here, x refers to the input function, e refers to the base of the natural logarithm (Euler’s number, almost equivalent to 2.71828), and $\text{Sigmoid}(x)$ is the output whose values range from zero to one.

$$\text{ReLU}(x) = \max(0, x) \quad (12)$$

The ReLU outputs the input values x if x is zero or positive and zero if

x is negative. In detail, it resembles a ramp, which allows positive values to pass through unchanged while changing negative values to 0.

$$L(y, \hat{y}) = -(y \cdot \log(\hat{y}) + (1 - y) \cdot \log(1 - \hat{y})) \quad (13)$$

In Eq. (13), $L(y, \hat{y})$ refers to the binary cross-entropy loss, y indicates the true label (0 or 1), and $\hat{y}$ shows the predicted probability belonging to class 1.

3.3.2. DBN classifier

The DBN comprises numerous layers of BP neural networks (BP-NN) and RBM [30]. RBM is a stochastic invention method fabricated from a hidden layer (HL) and visible layer (VL). The nodes aren’t linked in a similar layer, and nodes among layers are fully connected (FC) to equivalent weight parameters. Then, the DBN input refers to the hydrogen attention data perceived by the sensor. The HL and VL are random exponential family distributions. Every state contains a definite energy. If every unit values are formed, then the energy function $E(v, h)$ is stated by mathematical formulation as given below:

$$E(v, h) = - \sum_{i=1}^n \frac{(v_i - a_i)^2}{2\sigma_i^2} - \sum_{j=1}^m \frac{b_j h_j}{2\sigma_j^2} - \sum_{i=1}^n \sum_{j=1}^m \frac{W_{ij} v_i h_j}{\sigma_i^2 \sigma_j^2} \quad (14)$$

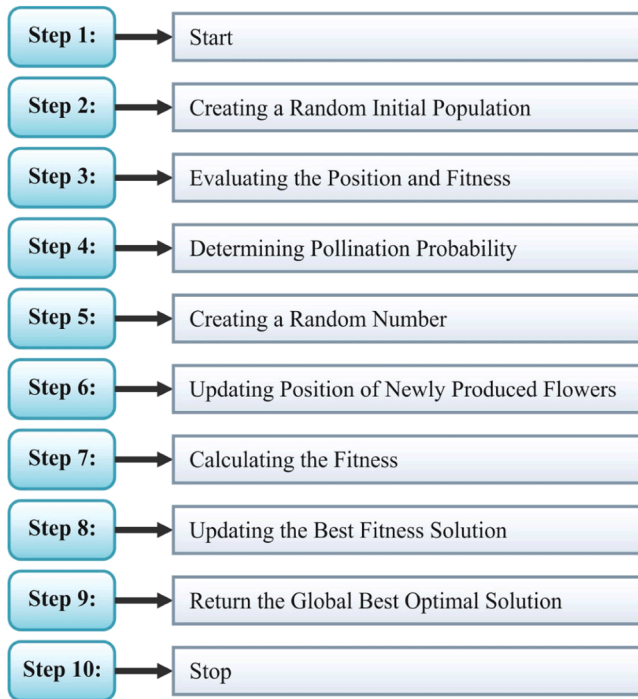


Fig. 6. Steps involved in the LEOA model.

Here, i signifies the i th neuron in the VL; j denotes the j th neuron in the HL; n and m indicate the count of neurons in the VL and the HL, correspondingly; W_{ij} epitomizes the weight from the i th and j th neurons, correspondingly; v_i and h_j denote the state of the i th and j th neurons, respectively; a_i and b_j embody the bias of the i th and j th neurons, respectively; σ_i and σ_j denote the standard deviation of neurons in the VL and HL, correspondingly.

The joint likelihood distribution is formulated below:

$$p(v, h) = \frac{\sum_n e^{-E(v,h)}}{\sum_{v,h} e^{-E(v,h)}} \quad (15)$$

When the VL state is assumed, the activation state of every HL is a probability distribution. In every layer, the conditional probability of neurons is conveyed as below:

$$p(h_j = h|v) = N\left(b_j + \sum_{i=1}^n \frac{W_{ij}v_i}{\sigma_i^2}, \sigma_j^2\right) \quad (16)$$

$$p(v_i = v|h) = N\left(a_i + \sum_{j=1}^m \frac{W_{ij}h_j}{\sigma_j^2}, \sigma_i^2\right) \quad (17)$$

Here, $N()$ denotes a Gaussian distribution.

The BP-NN is a multilayer feedforward NN that uses the error backpropagation (BP) method to enhance bias and weight parameters. It is made of single input and output layers and many HLs. Every layer covers numerous neural resources. In the structure of DBN, dual BP-NNs are harvested together, and the previous HL is captured as an input. The last output outcomes are the coordinate locations of the leakage sources X and Y correspondingly. The forward propagation procedure of BP-NN is stated below:

$$h_j^l = f\left(\sum_{i=1}^{s_{l-1}} \omega_{ij}^l h_i^{l-1} + b_j^l\right) \quad (18)$$

Here, h_j^l signifies the output of j th neuron in l th layer; h_i^{l-1} embodies the output of the i th neuron in $l-1$ th layer; ω_{ij}^l signifies the connection weight among the i th neuron in the $l-1$ th layer and j th neuron in l th

layer; b_j^l denotes the bias of the j th neuron in l th layer; s_{l-1} represents the highest neuron count in $l-1$ th layer; $f()$ epitomizes the non-linear activation function of l th layer. Fig. 5 depicts the structure of the DBN technique.

In the context of the amount of BP-NN layers in DBN and the neuron count, the loss function affected by the count of neurons and layers is frequently attuned.

For X and Y coordinate position outcomes by BP-NN, the weight parameter α is fixed, and the last outputs were upgraded as per the past position outcomes, as below:

$$y_{out}^t = y_{out}^{(t-1)} + \alpha(y_{out}^{(t-1)} - y^t) \quad (19)$$

Here, y_{out}^t refers to the position outcome by DBN in t second; y_{out}^{t-1} denotes the position outcome of DBN output at $t-1$ th second; y^t refers to the position outcome of BP-NN output at t th second.

3.4. Parameter optimizer

Eventually, the hyperparameter selection of the DL techniques arises using the LEOA model. The step-wise processing is enclosed to utilize the LEOA model to attain the finest values [31]. The LEOA model is appropriate for hyperparameter selection in DL methods due to its unique optimization capabilities inspired by the self-cleaning properties of lotus leaves. The LEOA method outperforms fine-tuning hyperparameters by effectively exploring the search space and averting local minima, often challenging conventional techniques. Its robustness and adaptability allow it to efficiently handle complex, high-dimensional parameter spaces. The emphasis on the LEOA model on balance effectually assists in attaining optimal hyperparameter configurations, resulting in enhanced model performance and mitigated computational overhead compared to other methods. This makes it a valuable tool for improving the efficiency of the DL methods in diverse and challenging datasets.

Utilizing the LEOA method for hyperparameter tuning can impact computational cost and time. While the LEOA method presents high-quality optimization by effectually exploring the parameter space and averting local minima, it may need significant computational resources due to its iterative behaviour and complex search strategies. Incorporating the LEOA method into existing DL model workflows can be feasible but might necessitate alterations in computational infrastructure to manage enhanced processing demands. The trade-off is balancing the high-quality outputs attained by LEOA against the potential growth in computational overhead, which should be considered based on the specific requirements and constraints of the application. Fig. 6 illustrates the steps involved in the LEOA model.

A new development model named Lotus Effect unites effective operators from the Dragonfly technique, water's self-cleaning capability for removal, and local search tasks. Initially, LEOA creates a population with an even distribution to enhance the parameters. LEOA is employed to uphold the finest choice.

Step1: Initialize

The weight parameter of the σ generator is employed to set the population of LEOA. The initial population of gas and its locations were stated and assumed in the mentioned calculation:

$$Z_i^r = -\sum_{j=1}^M Y_i^r - Y_j^r \quad (20)$$

Whereas Y_i signifies the current individual's location over index i in progress iteration r , Y_i denotes the j th distinct neighbourhood location in progress iteration r , and M represents the neighbourhood individuals' count.

Step2: Random Generation

The step vector is assumed to contain an effect on the LEOA throughout random establishment. The location vectors are considered as follows:

Table 1
Details of the dataset.

Type of Event	Data Record
Backdoor	5000
DoS	5000
DDoS	5000
Injection	5000
MITM	1000
Scanning	5000
Ransomware	5000
Password	5000
XSS	5000
Normal	5000
Total Record	46000

$$Y_i^{(r+1)} = Y_i^r + w\Delta Y_i^{(r+1)} \quad (21)$$

Whereas r signifies iteration count. Y_i indicates the individual's current position with index i in progress iteration r .

Step3: Fitness Function

The arbitrary solution is formed from the set of assessments. Assessment of fitness function (FF) considering outcomes of parameter optimizer for the originator's parameter of weight σ . Therefore, it is assumed as:

$$\text{Fitness function} = \text{Optimizing}[\sigma] \quad (22)$$

Step4: Exploration Stage

Once a new orientation is effectively defined, the LEOA can generate

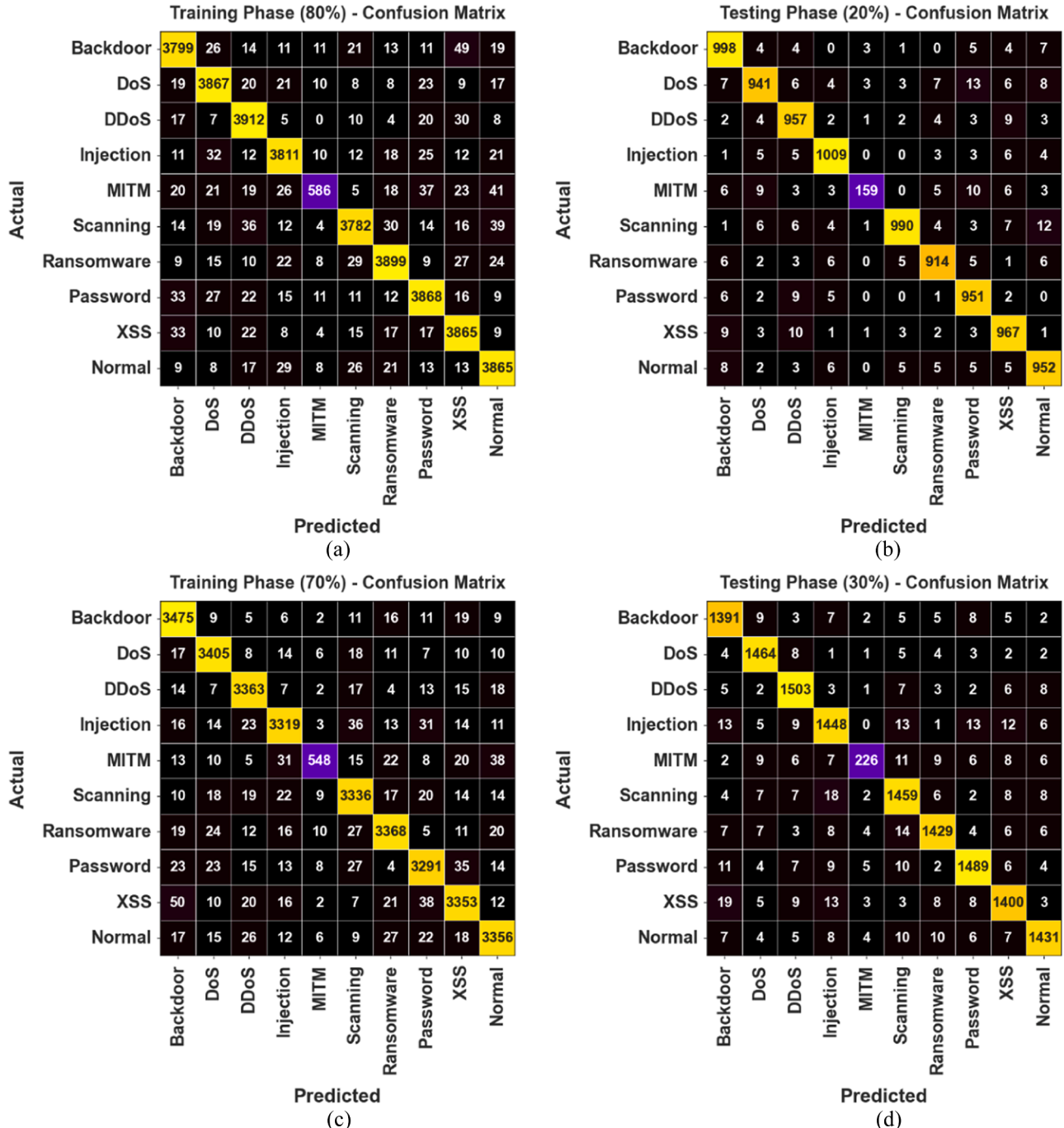


Fig. 7. Confusion matrices of (a-c) TRAP of 80 % and 70 % and (b-d) TESP of 20 % and 30 %.

Table 2

Attack detection outcome of NGCAD-EDLM technique under 80 %TRAP and 20 %TESP.

Classes	Accu _y	Prec _n	Rec _{a_i}	F1 _{Score}	MCC
TRAP (80 %)					
Backdoor	99.08	95.84	95.60	95.72	95.20
DoS	99.18	95.91	96.63	96.27	95.81
DDoS	99.26	95.79	97.48	96.63	96.22
Injection	99.18	96.24	96.14	96.19	95.73
MITM	99.25	89.88	73.62	80.94	80.98
Scanning	99.13	96.50	95.36	95.93	95.44
Ransomware	99.20	96.51	96.22	96.37	95.92
Password	99.12	95.81	96.12	95.97	95.47
XSS	99.10	95.20	96.63	95.91	95.41
Normal	99.10	95.38	96.41	95.89	95.39
Average	99.16	95.31	94.02	94.58	94.16
TESP (20 %)					
Backdoor	99.20	95.59	97.27	96.43	95.98
DoS	98.98	96.22	94.29	95.24	94.68
DDoS	99.14	95.13	96.96	96.04	95.56
Injection	99.37	97.02	97.39	97.21	96.85
MITM	99.41	94.64	77.94	85.48	85.61
Scanning	99.32	98.12	95.74	96.92	96.54
Ransomware	99.29	96.72	96.41	96.57	96.17
Password	99.18	95.00	97.44	96.21	95.76
XSS	99.14	95.46	96.70	96.08	95.60
Normal	99.10	95.58	96.06	95.82	95.32
Average	99.21	95.95	94.62	95.20	94.81

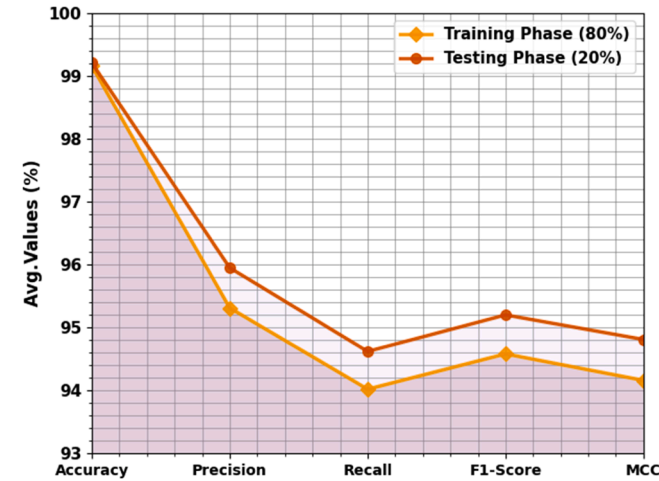


Fig. 8. Average of NGCAD-EDLM method under 80 %TRAP and 20 %TESP.

a series of numbers to discover searching space, employing them in significant formulation quite than randomly generated statistics, which increases the performance of exploration and expresses as:

$$Y_i^{(r+1)} = Y_i^r + T(Y_i^r - p^*) \quad (23)$$

Whereas Y_{r+1} represents pollen position in $(r + 1)^{th}$ iteration, p^* specifies the idea of originating pollen location among progress iterations, and T signifies the development of region and reduction model iteration. When the model meets the movement phases, the finest solutions are lengthier at the start and smaller as they travel near the conclusion. The calculation is expressed below:

$$T = 2e^{-\left(\frac{4r}{L}\right)^2} \quad (24)$$

Whereas r designates the algorithm's current progress iteration, L shows the last iteration count.

Step5: Exploitation Stage to Optimize σ

Table 3

Attack detection outcome of NGCAD-EDLM method under 70 %TRAP and 30 %TESP.

Classes	Accu _y	Prec _n	Rec _{a_i}	F1 _{score}	MCC
TRAP (70 %)					
Backdoor	99.17	95.10	97.53	96.30	95.84
DoS	99.28	96.32	97.12	96.72	96.32
DDoS	99.29	96.20	97.20	96.69	96.29
Injection	99.07	96.04	95.37	95.70	95.19
MITM	99.35	91.95	77.18	83.92	83.92
Scanning	99.04	95.23	95.89	95.56	95.02
Ransomware	99.13	96.15	95.90	96.02	95.54
Password	99.02	95.50	95.31	95.41	94.85
XSS	98.97	95.55	95.01	95.28	94.70
Normal	99.07	95.83	95.67	95.75	95.23
Average	99.14	95.39	94.22	94.74	94.29
TESP (30 %)					
Backdoor	99.14	95.08	96.80	95.93	95.46
DoS	99.41	96.57	97.99	97.28	96.95
DDoS	99.32	96.35	97.60	96.97	96.59
Injection	98.94	95.14	95.26	95.20	94.61
MITM	99.38	91.13	77.93	84.01	83.97
Scanning	98.99	94.93	95.92	95.42	94.85
Ransomware	99.22	96.75	96.03	96.39	95.96
Password	99.20	96.63	96.25	96.44	95.99
XSS	99.05	95.89	95.17	95.53	95.00
Normal	99.23	96.95	95.91	96.43	96.00
Average	99.19	95.54	94.49	94.96	94.54

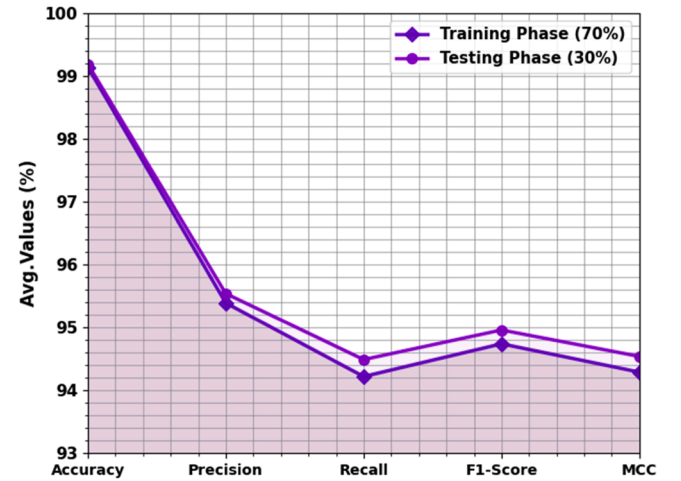


Fig. 9. Average of NGCAD-EDLM method under 70 %TRAP and 30 %TESP.

This stage is last, where the security tester intensely captures the benefits of faults that are being inspected. LEOA's exploitation stage is assumed below:

$$b_i^r = \frac{|k_i^r - k_{Max}| \times const}{(|k_{Min} - k_{Max}|)} \quad (25)$$

Whereas b_i^r signifies the ability of pit i in progress iteration r , k_i^r specifies the dimension of i^{th} in progress iteration r , k_{Max} shows the dimension of largest fitness between the pits, and k_{Min} is the dimension of the least fitness amid pits. The probability of selecting a pit among current is stated in the given calculation below:

$$Select_i^r = \frac{b_i^r}{\sum_{j=0}^t b_j^r} \quad (26)$$

Here, $Select_i^r$ denotes the probability of picking a pit, b_i^r specifies the ability of i^{th} pit on iteration r , and t denotes the sums of the pit whose size is larger than the overflowed pit.

Step6: End

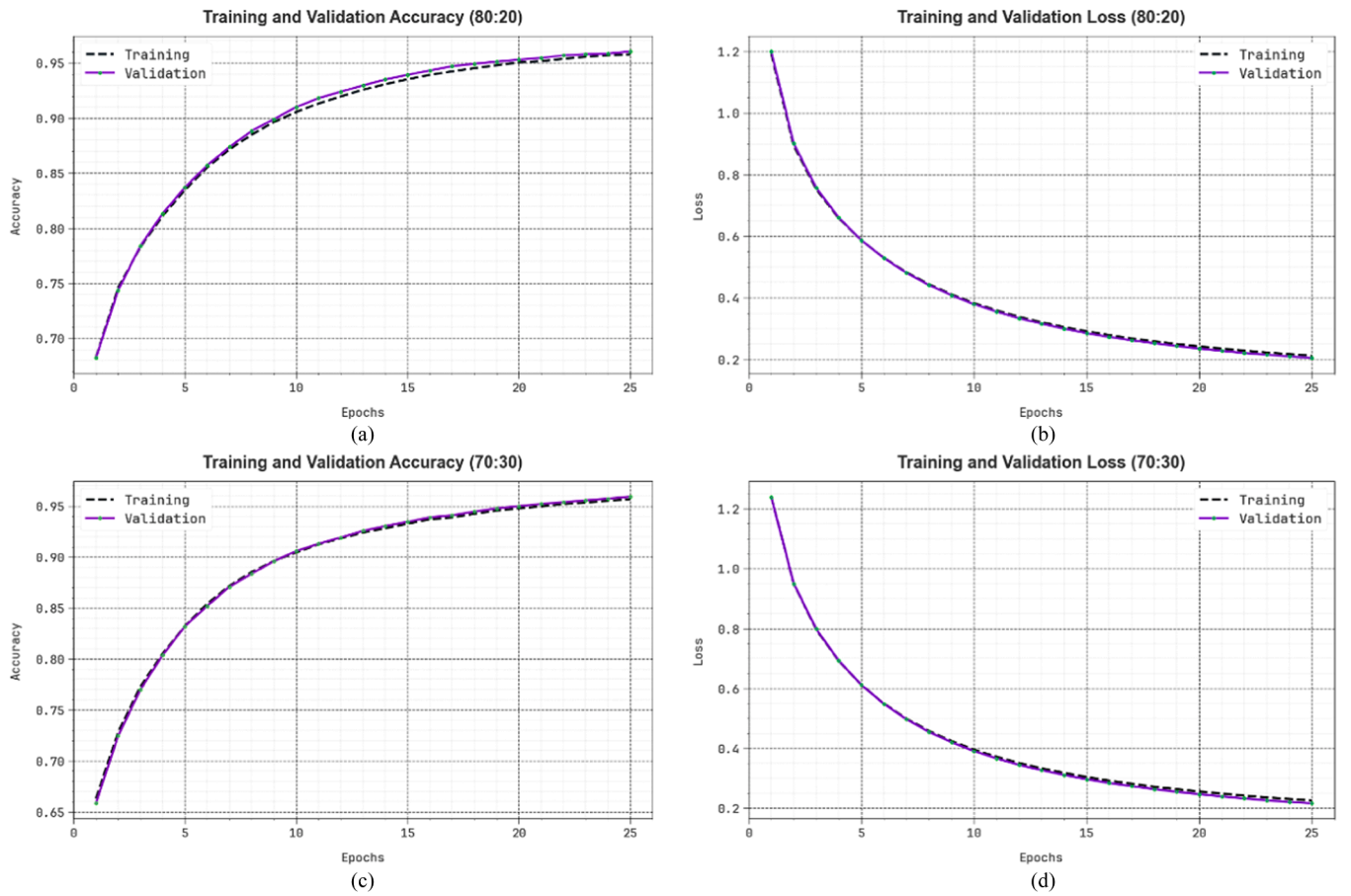


Fig. 10. (a-c) Accuracy of 80 %:20 % and 70 %:30 % and (b-d) Loss of 80 %:20 % and 70 %:30 %.

The weight parameter σ for enhanced utilizing LEOA will repeat the procedure till it fulfils the condition of halting $b = b + 1$. Lastly, AP-CNN-BTA enhances aesthetic capability and all-around growth of human units with superior precision by declining the computation and error time.

The LEOA develops an FF to achieve an enhanced classifier solution. It describes an optimistic integer that presents the superior performance of the candidate solution. In this research, the minimizer of the classifier rate of error is examined as FF and set in Eq. (27).

$$\text{fitness}(x_i) = \text{ClassifierErrorRate}(x_i)$$

$$= \frac{\text{number of misclassified samples}}{\text{Total number of samples}} \times 100 \quad (27)$$

4. Experimental validation

During this part, the experimental validation study of the NGCAD-EDLM method can be studied using the TON-IoT database [32], which comprises 46000 instances with ten classes signified in Table 1.

Fig. 7 depicted the confusion matrices generated by the NGCAD-EDLM method at 80:20 and 70:30 of TRAP/TESP. These results show that the NGCAD-EDLM model efficiently classified and recognized every 10 class labels.

Table 2 and Fig. 8 depict complete attack recognition results of the NGCAD-EDLM method under 80 %TRAP and 20 %TESP. The results show that the NGCAD-EDLM method has properly identified and classified ten classes. With 80 %TRAP, the NGCAD-EDLM methodology offers an average $accu_y$ of 99.16 %, $prec_n$ of 95.31 %, $reca_l$ of 94.02 %, $F1_{score}$ of 94.58 %, and MCC of 94.16 %. In addition, with 20 %TESP, the NGCAD-EDLM methodology provides an average $accu_y$ of 99.21 %, $prec_n$

of 95.95 %, $reca_l$ of 94.62 %, $F1_{score}$ of 95.20 %, and MCC of 94.81 %.

Table 3 and Fig. 9 portray a complete attack recognition outcome of the NGCAD-EDLM approach under 70 %TRAP and 30 %TESP. This outcome is implicit that the NGCAD-EDLM approach has appropriately classified and identified 10 class labels. With 80 %TRAP, the NGCAD-EDLM technique gives an average $accu_y$ of 99.14 %, $prec_n$ of 95.39 %, $reca_l$ of 94.22 %, $F1_{score}$ of 94.74 %, and MCC of 94.29 %. Additionally, with 30 %TESP, the NGCAD-EDLM technique presents an average $accu_y$ of 99.19 %, $prec_n$ of 95.54 %, $reca_l$ of 94.49 %, $F1_{score}$ of 94.96 %, and MCC of 94.54 %.

Fig. 10 stated the classifier outcomes of the NGCAD-EDLM approach at 80 %:20 % and 70 %:30 %. Figs. 10a-10c depicts the accuracy study of the NGCAD-EDLM approach. The figure shows that the NGCAD-EDLM method achieves increased $accu_y$ performance over growing epoch counts. The progressive validation $accu_y$ overtraining $accu_y$ shows that the NGCAD-EDLM model learns proficiently of the test database. Finally, Figs. 10b-10d exhibits the loss study of the NGCAD-EDLM technique. These results indicate that the NGCAD-EDLM approach achieves adjacent training and validation loss outcomes. It is detected that the NGCAD-EDLM approach studies the test datasets well.

Fig. 11 indicated the classifier outcomes of the NGCAD-EDLM model at 80 %:20 % and 70 %:30 %. Figs. 11a-11c illustrates the PR study of the NGCAD-EDLM method. The values reported that the NGCAD-EDLM model outcomes in increased performance of PR. In addition, the NGCAD-EDLM method can attain maximum PR values on 10 class labels. At last, Figs. 11b-11d showed the analysis of ROC of the NGCAD-EDLM method. This figure denotes that the NGCAD-EDLM approach led to better values for ROC. Correspondingly, it is clear that the NGCAD-EDLM model can lengthen improved values of ROC on ten classes.

In Table 4 and Fig. 12, the simulation outcomes of the NGCAD-EDLM

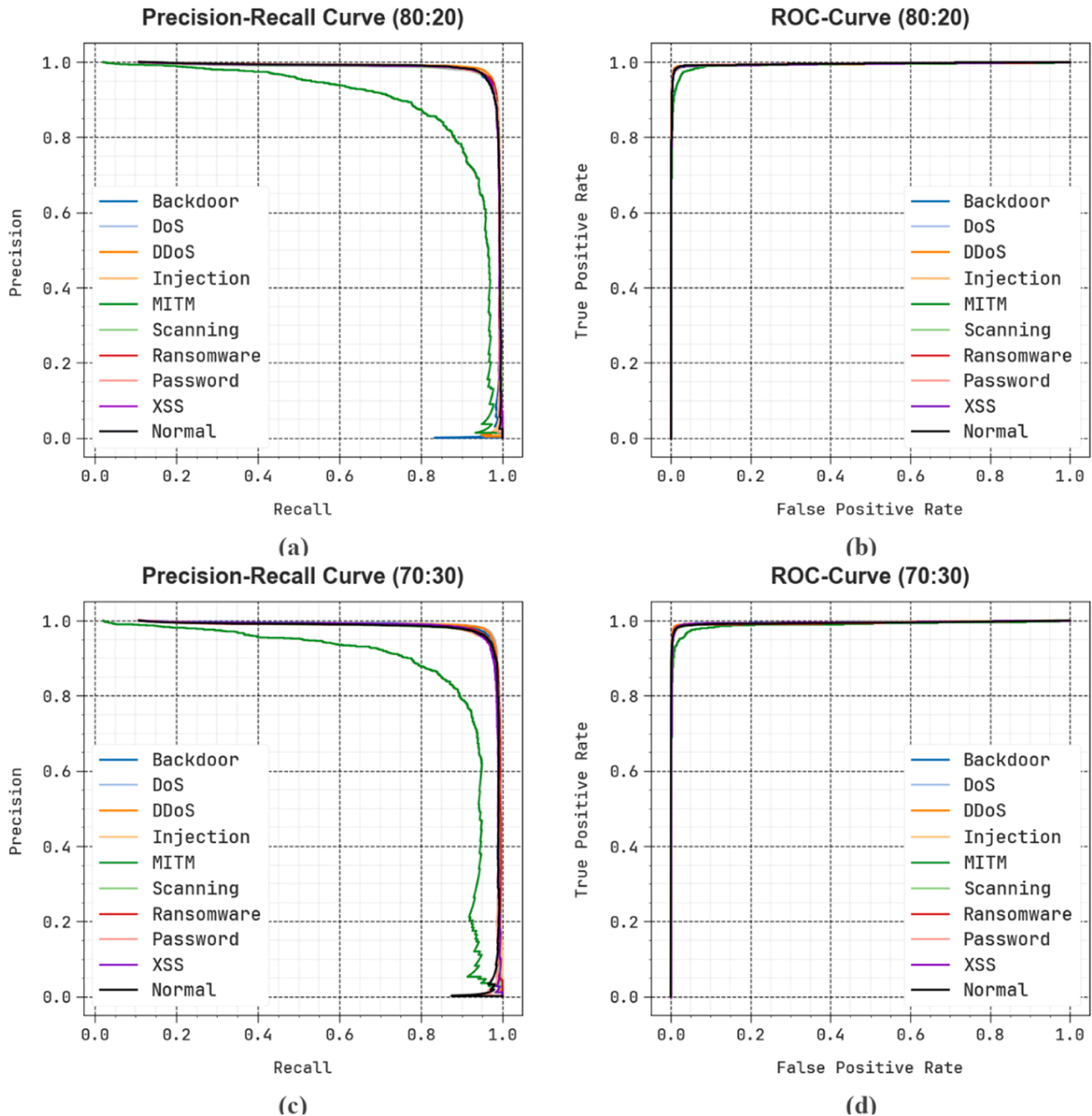


Fig. 11. (a-c) PR of 80 %:20 % and 70 %:30 % and (b-d) ROC of 80 %:20 % and 70 %:30 %.

Table 4
Comparative study of NGCAD-EDLM approach with existing models [33–35].

Technique	$Accu_y$	$Prec_n$	$Reca_l$	$F1_{score}$
DenseNet Classifier	98.84	91.22	93.35	94.68
NB	98.57	93.75	94.04	92.83
XGBoost Classifier	98.30	92.06	92.70	94.45
Contextual information	96.10	92.83	91.02	93.41
CSG	96.78	90.48	94.15	93.61
LSTM	92.29	91.25	92.37	93.77
MLP	88.85	93.13	93.08	93.25
NGCAD-EDLM	99.21	95.95	94.62	95.20

method with recent techniques such as DenseNet Classifier, Naïve Bayes (NB), XGBoost Classifier, Contextual information, Cyber Security Game (CSG), LSTM, and MLP models are provided [33–35]. The outcomes

show that the MLP technique has presented lower performance with $accu_y$, $prec_n$, $reca_l$, and $F1_{score}$ of 88.85 %, 93.13 %, 93.08 %, and 93.25 %, correspondingly. Simultaneously, the LSTM approach has achieved a slightly enhanced results with $accu_y$, $prec_n$, $reca_l$, and $F1_{score}$ of 92.89 %, 91.25 %, 92.37 %, and 93.77 %, appropriately. Also, the NB, XGBoost, Contextual information and CSG techniques have achieved reasonably near performance. In the meantime, the DenseNet approach has resulted in significant conclusions with $accu_y$, $prec_n$, $reca_l$, and $F1_{score}$ of 98.84 %, 91.22 %, 93.35 %, and 94.68 %, correspondingly. Then the NGCAD-EDLM model exceeds another method with greater $accu_{racy}$, $prec_n$, $reca_l$, and $F1_{score}$ of 99.21 %, 95.95 %, 94.62 %, and 95.40 %, appropriately.

Table 5 and Fig. 13 show the computational time (CT) analysis of the NGCAD-EDLM method with the recent method. The NGCAD-EDLM method achieves the lowest processing time of 0.67 seconds. In

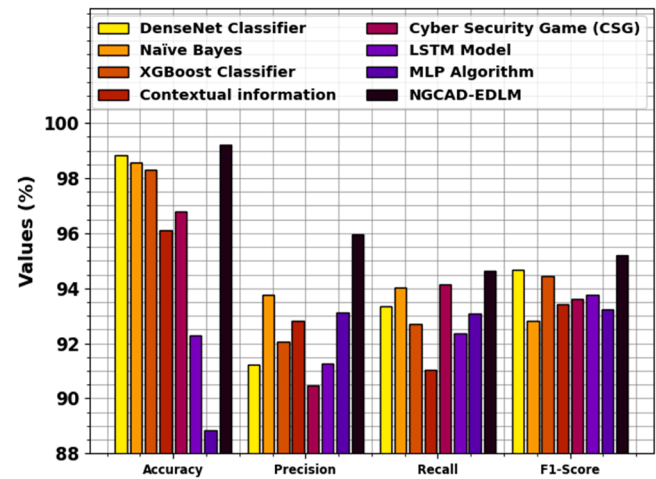


Fig. 12. Comparative analysis of NGCAD-EDLM technique with recent models.

Table 5
CT evaluation of NGCAD-EDLM approach with existing methods.

Technique	CT (sec)
DenseNet Classifier	3.79
NB	2.08
XGBoost Classifier	2.01
Contextual information	3.03
CSG	1.54
LSTM	5.24
MLP	5.08
NGCAD-EDLM	0.67

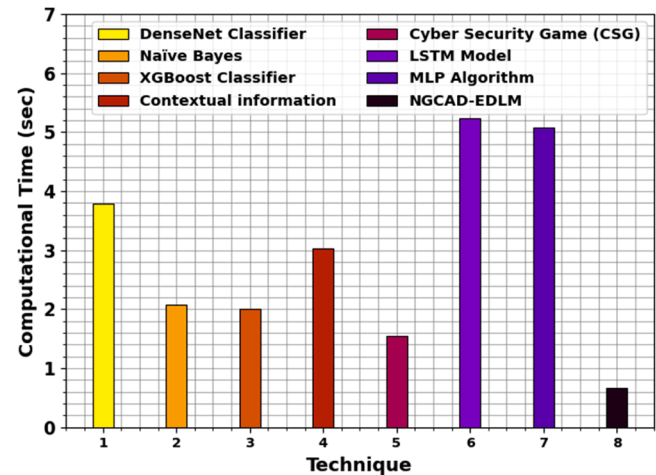


Fig. 13. CT evaluation of NGCAD-EDLM approach with existing methods.

comparison, the DenseNet Classifier takes 3.79 seconds, Naïve Bayes requires 2.08 seconds, and the XGBoost Classifier needs 2.01 seconds. Other techniques, namely Contextual Information and CSG, have times of 3.03 and 1.54 seconds. LSTM and MLP methodologies have the highest computational times at 5.24 and 5.08 seconds. This demonstrates that the NGCAD-EDLM technique is significantly faster than the other methods. Its superior speed can substantially reduce operational delays and enhance overall system performance, making it specifically advantageous for environments needing rapid data processing and immediate response.

5. Conclusion

In this paper, an NGCAD-EDLM technique is proposed in the IIoT environment. The foremost cause of the NGCAD-EDLM technique is the automated recognition of cyber-attacks. The NGCAD-EDLM approach has data normalization, HBA-based feature selection, EL, and parameter optimizer to accomplish that. Initially, the NGCAD-EDLM approach undergoes data normalization utilizing min-max normalization has been carried out. Next, the HBA is used to select feature sub-sets. Furthermore, an ensemble DL of two methods, namely CNN and DBN models, are applied. Eventually, the hyperparameter selection of the DL techniques arises using the LEOA method. A complete set of simulation validation is conducted to establish the experimental analysis of the NGCAD-EDLM method. The empirical findings reported a better outcome of the NGCAD-EDLM model over other present approaches. Incorporating CNN and DBN models for cybersecurity attack detection in IIoT systems can encounter limitations such as high computational costs and complexity in model training. The system may also face difficulty with scalability and adaptability to various and growing attack patterns. Future studies should concentrate on improving the effectualness of the incorporated model through more optimized training algorithms and mitigating computational overhead. Furthermore, developing methods to enhance the capability of the technique to generalize across diverse attack scenarios and IoT environments would be beneficial. Research into hybrid techniques that integrate these models with real-time detection capabilities could advance the efficiency and practicality of the model.

CRedit authorship contribution statement

Mahmoud Ragab: Project administration, Methodology, Formal analysis, Conceptualization. **Nasser N. Albogami:** Writing – original draft, Validation, Resources, Investigation. **Omar A. Abdulkader:** Writing – original draft, Resources, Formal analysis. **Hashem Alaidaros:** Writing – review & editing, Validation, Software. **Mohammed Basher:** Visualization, Funding acquisition, Data curation, Conceptualization. **Alanoud Subahi:** Writing – original draft, Software, Investigation, Formal analysis. **Hassan Mousa:** Writing – review & editing, Writing – original draft, Visualization. **Abdullah AL-Malaise AL-Ghamdi:** Resources, Conceptualization.

Declaration of Competing Interest

All authors were fully involved in the preparation and study of the manuscript, and we all have read and concur with its contents. The authors have no conflict of interest. This manuscript has not been published elsewhere and are not currently under consideration by another journal published by You or any other publisher. We would appreciate considering this paper for publication and we have a full agreement to pay the publishing cost as determined by the journal management

Data Availability

The data supporting this study’s findings are available in the Kaggle repository at <https://research.unsw.edu.au/projects/toniot-datasets>, reference number [32].

Acknowledgements

This Project was funded by the Deanship of Scientific Research (DSR) at King Abdulaziz University (KAU), Jeddah, Saudi Arabia, under grant no. (GPIP: 1307-611-2024). Therefore, the authors acknowledge with thanks DSR at KAU for technical and financial support.

Institutional review board statement

Not Applicable.

Informed consent statement

Not Applicable.

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